SCIENCE 2: BOARD MODEL EXAMINATION & QUESTION BANK

Ref: SCI2-EXAM-2026 | Academic Year 2025–2026 | Class 10 (SSC)

SECTION A: Objective Questions & Knowledge Retrieval

Time Allowed: 2 Hours | Maximum Marks: 40 | Question 1 (A & B) - 10 Marks

General Instructions for Examination

1. All questions are compulsory. Internal choices are provided within respective sections.
2. Draw neat, well-labelled diagrams wherever required in biology questions.
3. Numbers to the right indicate full marks for that specific question.
4. Evaluation follows the updated Maharashtra State Board Class 10 Science 2 Blueprint.

Q.1 (A) Choose the correct alternative and complete statements (5 Marks)

- 1. The transfer of genetic information from a molecule of DNA to mRNA is called as _____ process. (a) Translation (b) Transcription (c) Translocation (d) Mutation [Bloom: Remember | Difficulty: Easy]
- *Ans: (b) Transcription - mRNA is synthesized complementary to the DNA template strand under RNA polymerase.*
- 2. Vestigial organ _____ present in the human body is morphological proof of evolution. (a) Appendix (b) Intestine (c) Lungs (d) Heart [Bloom: Remember | Difficulty: Easy]
- *Ans: (a) Appendix - Appendix is functionless in humans but functional in ruminant herbivores for cellulose digestion.*
- 3. The causality behind sudden evolutionary genetic variations was explained by _____ through his Mutation Theory. (a) Hugo de Vries (b) Gregor Mendel (c) Charles Darwin (d) Lamarck [Bloom: Remember | Difficulty: Easy]
- *Ans: (a) Hugo de Vries (1901) formulated the Mutation theory of evolution.*
- 4. Organisms that produce their own organic food using sunlight and chlorophyll are called _____. (a) Producers (b) Primary consumers (c) Decomposers (d) Parasites [Bloom: Remember | Difficulty: Easy]
- *Ans: (a) Producers (Autotrophs).*
- 5. Electricity generated by using thermal energy from solar radiation is known as _____ electricity. (a) Solar thermal (b) Nuclear (c) Hydroelectric (d) Geothermal [Bloom: Remember | Difficulty: Easy]
- *Ans: (a) Solar thermal electricity generation.*

SECTION A (Contd.): Correlation, Odd One Out & Definitions

Question 1 (B) - 5 Marks | Factual & Conceptual Recall

Q.1 (B) Solve the following sub-questions (1 Mark Each)

- 1. Find the odd one out and justify: Transcription, Translation, Translocation, Mutation. [Bloom: Analyze | Difficulty: Easy]
- *Ans: Mutation. Reason: Transcription, translation, and translocation are normal sequential biological steps of protein synthesis, whereas mutation is an accidental sudden change in nucleotide sequence.*
- 2. Identify correlation: Peripatus : Connecting link :: Human appendix : _____ [Bloom: Understand | Difficulty: Easy]
- *Ans: Vestigial organ.*
- 3. State True or False with rectification: "The gender of an unborn child is scientifically determined by the maternal ovum." [Bloom: Evaluate | Difficulty: Medium]
- *Ans: False. Rectification: The gender of a human child is determined by the paternal sperm (which carries either an X or a Y chromosome); all maternal ova contain only X chromosomes.*
- 4. Write the name of the international disaster relief and search agency in India: [Bloom: Remember | Difficulty: Easy]
- *Ans: National Disaster Response Force (NDRF).*
- 5. Match the following: (Column A: Solar panel) -> (Column B: Conversion): [Bloom: Understand | Difficulty: Easy]
- *Ans: Photovoltaic cells convert radiant solar energy directly into direct current (DC) electrical energy without rotating turbines.*

SECTION B: Conceptual Understanding & Reasoning (Q.2 - 10 Marks)

Attempt any FIVE out of the following seven sub-questions (2 Marks each).

- Q.2 (1) Give scientific reason: Peripatus is designated as a living connecting link between Phylum Annelida and Phylum Arthropoda. [Bloom: Understand | Difficulty: Medium]
- *Ans: Peripatus displays segmented body, thin cuticle, and parapodia-like organ structures characteristic of Annelids, while simultaneously exhibiting trachea and an open circulatory system typical of Arthropods.*
- Q.2 (2) Give scientific reason: Microbial enzymes are preferred over chemical catalysts in modern eco-friendly industries. [Bloom: Analyze | Difficulty: Medium]
- *Ans: Microbial enzymes function effectively at lower temperatures, normal atmospheric pressure, and neutral pH, thus drastically saving thermal energy, eliminating hazardous effluents, and providing higher substrate specificity.*

SECTION B (Contd.) & SECTION C: Analytical Reasoning

Question 2 (Contd.) & Question 3 - Short & Intermediate Analytical Questions

Q.2 (Contd.) Attempt Remaining Questions (2 Marks Each)

- Q.2 (3) Distinguish between: Lamarckism vs Darwinism (Natural Selection). [Bloom: Analyze | Difficulty: Medium]
- *Ans: Lamarckism proposes that characters acquired by an organism through continuous use or disuse during its lifetime are transmitted to offspring. Darwinism states that nature naturally selects pre-existing favorable genetic variations that confer higher survival fitness.*
- Q.2 (4) Explain two major functions of the testis in the human male reproductive system. [Bloom: Understand | Difficulty: Easy]
- *Ans: 1. Production and maturation of male haploid gametes (sperms) inside seminiferous tubules. 2. Secretion of the steroid hormone testosterone responsible for secondary male sexual characteristics.*
- Q.2 (5) Define vestigial organs. List any four vestigial organs found in the human body. [Bloom: Remember | Difficulty: Easy]
- *Ans: Vestigial organs are degenerated or underdeveloped useless organs of organisms. Examples: 1. Appendix 2. Tail-bone (Coccyx) 3. Wisdom teeth 4. Ear muscles (Pinna movement).*

SECTION C: Detailed Explanations & Justifications (Q.3 - 15 Marks)

Attempt any FIVE out of seven questions (3 Marks each).

- Q.3 (1) Read the statement and justify in your own words: "Geographical and reproductive isolation of organisms gradually leads to speciation." [Bloom: Evaluate | Difficulty: Hard]
- *Ans: Each species thrives in distinct ecological habitats with unique dietary, climatic, and reproductive requirements. When a population becomes isolated by geographical barriers (mountains, rivers, continental drift), gene flow ceases. Over extended evolutionary epochs, accumulated genetic mutations and varied natural selection pressures create reproductive incompatibility, establishing a distinct new species.*
- Q.3 (2) Justify why the study of fossils and carbon-dating methods is indispensable in evolutionary biology. [Bloom: Evaluate | Difficulty: Medium]
- *Ans: Fossils preserve structural remnants and impressions of ancient organisms across geological rock strata. Radiocarbon dating (measuring remaining C-14 compared to stable C-12) establishes precise chronology, proving that invertebrates evolved first in ancient eras, followed by fish, amphibians, reptiles, and mammals.*

SECTION C (Contd.): Applied Disasters & Environmental Health

Question 3 (Contd.) - 3 Marks Each | Practical Application & Bloom: Apply

Q.3 (Contd.) Practical Science 2 Case Studies

- Q.3 (3) What are the core objectives of conducting a disaster management mock drill in schools and public centers? [Bloom: Apply | Difficulty: Medium]
- *Ans: 1. To evaluate readiness, quick response time, and operational coordination during sudden emergencies. 2. To test the functionality of firefighting gear, alarms, and emergency evacuation exits. 3. To impart hands-on rescue and first-aid skills, eliminating panic among students and personnel.*
- Q.3 (4) Explain the three primary stages of the Disaster Management Cycle: Pre-disaster, Emergency, and Post-disaster recovery. [Bloom: Understand | Difficulty: Medium]
- *Ans: • Pre-disaster: Hazard mapping, early warning systems, preventive infrastructure reinforcement, and citizen training. • Emergency Phase: Immediate search and rescue, medical triage, food/water supply, and life preservation. • Post-disaster: Debris removal, disease prevention, infrastructure rehabilitation, and long-term socio-economic rebuilding.*
- Q.3 (5) Write short notes on: Cybercrime hazards and technology addiction among teenagers. [Bloom: Understand | Difficulty: Medium]
- *Ans: Excessive screen time and social media fixation cause interpersonal alienation, visual strain, insomnia, and psychological anxiety. Cyber risks include phishing fraud, unauthorized biometric sharing, and cyberbullying. Remedial measures involve parental counseling, digital detox habits, sports participation, and statutory reporting under the IT Act.*
- Q.3 (6) Distinguish between: Conventional Thermal Power Generation vs Solar Photovoltaic Generation. [Bloom: Analyze | Difficulty: Medium]
- *Ans: Thermal generation burns coal to generate steam, emitting harmful greenhouse gases (CO2, SO2) and depleting fossil reserves. Solar PV generation directly converts solar radiation into DC voltage without mechanical boilers, generating clean, renewable, and pollution-free power.*

SECTION D: Long Answer & Comprehensive Synthesis

Question 4 - 5 Marks | Deep Explanations, Central Dogma & Question Bank Matrix

Q.4 Attempt any ONE of the following comprehensive questions (5 Marks)

- Q.4 (1) Explain the Central Dogma of Protein Synthesis in living organisms. Detail the three coordinated stages: Transcription, Translation, and Translocation with appropriate terminology. [Bloom: Create / Understand | Difficulty: Hard]
- *Ans: The Central Dogma establishes that genetic information flows unidirectionally from DNA -> RNA -> Functional Protein.*
- • Transcription: Inside the cell nucleus, the double-stranded DNA uncoils. The enzyme RNA polymerase synthesizes a single-stranded messenger RNA (mRNA) complementary to the DNA template strand, replacing Thymine (T) with Uracil (U).
- • Translation: The synthesized mRNA emerges into the cytoplasm and docks onto a ribosome. The mRNA sequence is deciphered in groups of three nitrogenous bases called triplet codons. Transfer RNA (tRNA) molecules possessing complementary anticodons transport specific amino acids to the ribosomal site.
- • Translocation: The ribosome sequentially advances along the mRNA chain by the distance of exactly one triplet codon. Peptide bonds are synthesized between successive amino acids, assembling a nascent polypeptide chain.
- Q.4 (2) Describe the flood disaster management framework across pre-disaster, warning, emergency rescue, rehabilitation, and long-term riverbank reconstruction phases. [Bloom: Evaluate | Difficulty: Hard]
- *Ans: Pre-disaster: River catchment desilting, radar meteorological monitoring, and flood-line zoning. Warning: Sirens, emergency SMS broadcasting, and livestock evacuation. Emergency: NDRF motorized boat deployment, helivac of stranded victims, water purification kits. Rehabilitation: Shelter camps, chlorination against cholera, and school resumption.*

Board Question Bank Specification & Marking Summary

Total Marks: 40 | Question Count: 18 Questions across 4 Sections.

Cognitive Distribution: Remember: 25% | Understand: 35% | Apply: 20% | Analyze: 12% | Evaluate/Create: 8%.

Difficulty Distribution: Easy: 40% (16 Marks) | Medium: 42% (17 Marks) | Hard: 18% (7 Marks).